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Tech News

Apple's New Screen Repair Trap Could Change the Repair Industry Forever



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Apple Broke iPhone 13 Screen Repairs



iFixit's Taylor Dixon digs into the process of removing the iPhone 13's Face-ID-saving screen chip, including an interview with The Art of Repair's Justin Ashford.

Update: Apple [told The Verge on Nov. 9](#), five days after this post was widely cited in news reports, that it would [issue a software update](#) to prevent Face ID's disabling after screen replacements.

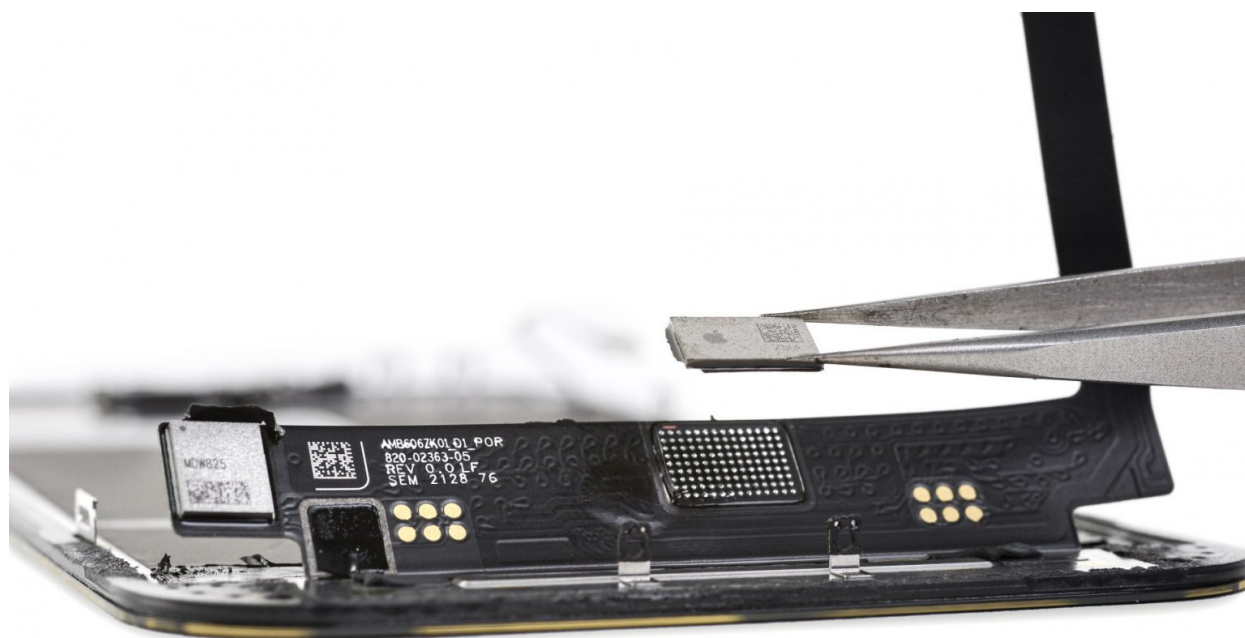
Apple has been chipping away at iPhone repair work outside their control for years now. With new changes to the iPhone 13, they may be aiming to shatter the market completely.

The new iPhone 13 completely disables its flagship Face ID functionality when you replace its screen. We have confirmed this repeatedly in our lab, testing with many different phones on iOS 15 and 15.1, and our results have been replicated by [numerous repair professionals](#).

This is a dark day for fixers, both DIY and professional. One of the most common phone repairs that could once be [done with hand tools](#) now requires a microscope. This means you won't be able to fix your [iPhone screen](#) yourself without sacrificing major functionality. It also has huge implications for the professional repair industry, for which Apple is the dominant brand to service. Small shops could be shuttered, forced to choose between spending thousands on new equipment or losing a major

source of income.

For shops that want to survive, their only options will be to join Apple's [onerous IRP network](#)—not an option for shops that value their customers' privacy—or work past the iPhone's locks with microsoldering tools and training. This unprecedented lockdown is unique to Apple. It's totally new in the iPhone 13, and hard to understand as a security measure, given that [the Face ID illuminator is entirely separate from the screen](#). It is likely the strongest case yet for right to repair laws. And it's all because of a chip about the size of a Tic-Tac, tucked into the bottom of a screen.



Removing the center chip from an iPhone 13 display.

The iPhone 13 is paired to its screen using this small microcontroller, in a condition repair techs often call “serialization.” Apple has not provided a way for owners or independent shops to pair a new screen. Authorized technicians with access to [proprietary software, Apple Services Toolkit 2, can make new screens work by logging the repair](#) to Apple's cloud servers and syncing the serial numbers of the phone and screen. This gives Apple the ability to approve or deny each individual repair.

iPhone 13 screen replacement disables Face ID in iOS15. Why? Our...



iCorrect shows one procedure for removing the iPhone 13's display, and verifies that it re-enables Face ID, once transplanted onto a new screen.

The most sophisticated repair shops have found a workaround, but it's not a quick, clever hack—it's physically moving a soldered chip from the original screen onto the replacement. We'll go into more detail on that process below, but it's important to note how completely unprecedented this is. Screen replacement is incredibly common. Tens of thousands of repair shops around the world support their communities by replacing screens for customers at competitive prices. And Apple is, with one fell swoop, seemingly cutting the industry off at the knees.

Justin Drake Carroll, CEO and founder of [Fruit Fixed](#), a regional repair chain in Virginia, said that screen replacements were about 35 percent of revenue. "At one point it was 60 percent, a few years ago. We worked really hard to push that figure down, so that one revenue stream wasn't such a huge part of what we do. Obviously, it's still an incredibly important part of our business model."